Frames and matrix designs

Zauner's conjecture on maximal complex equiangular tight frames

Difficulty: extreme **Importance:** broadly interesting

Status: Partially resolved

Rating rationale: All-dimension SIC existence is a central structured-design barrier connecting frame theory, quantum information and arithmetic geometry.

For every integer $d \geq 2$, do there exist d^2 unit vectors $\phi_1, \ldots, \phi_{d^2} \in \mathbb{C}^d$ satisfying

$$|\phi_i^* \phi_j|^2 = \frac{1}{d+1} \quad (i \neq j), \qquad \sum_{i=1}^{d^2} \phi_i \phi_i^* = dI_d?$$

This is existence of an equiangular tight frame with d^2 vectors in $\mathbb{C}^d$, equivalently a symmetric informationally complete positive operator-valued measure after dividing the rank-one projectors by d. The statement concerns existence in every dimension. It does not impose additional Weyl–Heisenberg or order-three symmetry.

These frames attain the optimal coherence for d^2 unit vectors and lead to highly symmetric Gram matrices. Their construction is a concrete structured matrix design and conditioning problem with applications to state reconstruction. Exact all-dimension existence remains substantially stronger than numerical constructions in many individual dimensions.

References

1. A. S. Bandeira et al., *Randomstrasse 101: Open Problems of 2025*, arXiv:2603.29571 (2026), Conjecture 24 and its definition of equiangular tight frames. Paper.
2. M. Appleby, S. T. Flammia, and G. S. Kopp, *A Constructive Approach to Zauner's Conjecture via the Stark Conjectures*, arXiv:2501.03970v2 (2025). The abstract and main construction distinguish conjectural/conditional arithmetic input from unconditional all-dimension existence. Paper.
3. S. Joka, *Symmetric Informationally Complete Positive Operator Valued Measure and Zauner conjecture*, arXiv:2601.13475v5. Cited only as an excluded proof claim: withdrawn May 31, 2026, with the author's comment that the proof is incorrect. Withdrawal record.

Status check — 2026-09-10

Searched "Zauner conjecture proof 2026", "SIC POVM all dimensions solved", and checked the latest versions rather than relying on search snippets. The apparent January 2026 proof claim is withdrawn; its unchanged abstract must not be read as a valid solution. The Appleby–Flammia–Kopp construction does not give an unconditional proof of the statement. No other general resolution was found.

Audit update (2026-09-10): Rechecked the Appleby–Flammia–Kopp conditional construction and Joka's withdrawn v5, and searched for other general proofs. Many individual dimensions have exact constructions, but the all-dimension existence question remains unresolved; the withdrawn claim is not a solution. This is a bounded literature check, not a proof that no solution exists.